

Simulation Study

Jambu Zoo Visitor Flow Simulation

A Monte Carlo simulation of zoo visitors based on real observation data



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Objective of the Project

Why we built this simulation

Goal

To simulate **visitor behaviour at Jambu Zoo** using probability distributions derived from real observed data and forecast operational metrics across a defined timeline.

Specific Aims

- Capture variability in visitor demographics & behavior
- Model arrival times, transport, payment & duration
- Generate 200+ simulated visitors using Monte Carlo
- Derive actionable inferences for zoo operations

200+

simulated visitors

9

input variables

63

observed records

2

days of data (Sat & Sun)

Variables Selected

Nine input variables drive the simulation

1

Age

Visitor age band (20–69)

Demographic

2

Gender

Male / Female

Demographic

3

Day

Saturday / Sunday

Temporal

4

Arrival Time

13:00 – 17:00 windows

Temporal

5

Accompanied By

Group size 1 - 6

Behavioural

6

Mode of Transport

Auto, Bike, Bus, Cab, Car

Behavioural

7

Payment Mode

Cash, Card, UPI

Transactional

8

Eateries Used

Yes / No

Behavioural

9

E-cart Service

Yes / No

Service

Data Collection & Range Tables

63 visitor records → probability ranges

Data Collection

Method: On-site observation at Jambu Zoo across Saturday & Sunday.

Sample: 63 visitors logged across 9 attributes.

Window: 13:00 – 17:00 (peak afternoon hours).

Age Distribution

Age Band	Probability
20–29	0.159
30–39	0.571
40–49	0.175
50–59	0.079
60–69	0.016

Gender

Value	Prob.	Cum.
Male	0.429	0.429
Female	0.571	1.000

Payment Mode

Value	Prob.	Cum.
Cash	0.476	0.476
Card	0.143	0.619
UPI	0.381	1.000

Mode of Transport

Value	Prob.	Cum.
Auto	0.111	0.111
Bike	0.032	0.143
Bus	0.381	0.524
Cab	0.127	0.651
Car	0.349	1.000

Visit Duration (min)

Value	Prob.	Cum.
0–60	0.016	0.016
60–89	0.317	0.333
90–119	0.302	0.635
120–149	0.127	0.762
150–179	0.127	0.889
180–209	0.111	1.000

Simulation for the Defined Timeline

Monte Carlo over 13:00 – 17:00

Method

- 1

Generate random number RAND() per variable
- 2

Map RAND to value via cumulative-probability table
- 3

Repeat for 200+ simulated visitors
- 4

Aggregate results to derive operational inferences

Sample Simulated Visitors

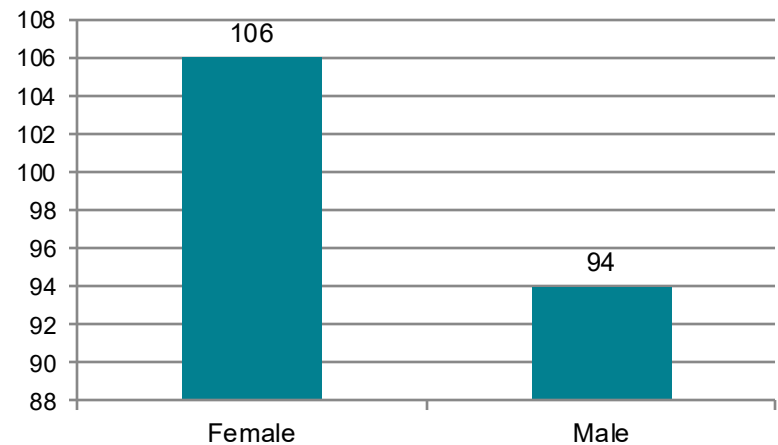
#	Age	Gender	Transport	Pay	Eat	Dur
1	50-59	Male	Bus	UPI	Yes	90-119
2	30-39	Female	Bus	UPI	Yes	90-119
3	40-49	Female	Bus	Cash	Yes	60-89
4	30-39	Male	Cab	UPI	Yes	90-119
5	20-29	Female	Bus	Cash	Yes	150-179
6	20-29	Male	Bus	Card	No	60-89
7	30-39	Female	Car	Cash	No	60-89
8	40-49	Male	Bus	UPI	Yes	60-89
9	40-49	Male	Bus	Card	Yes	180-209
10	30-39	Female	Auto	Cash	Yes	60-89

... 200+ visitors simulated in total

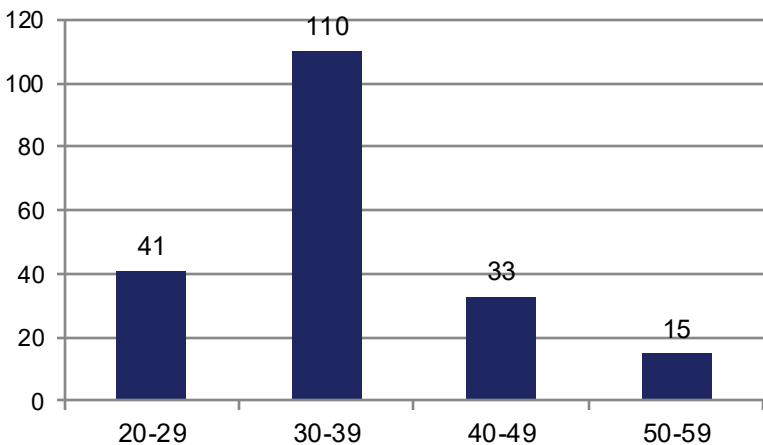
Inferences from Simulation (Part 1)

Demographics & Visit Behaviour

Gender Split (n=200)



Age Distribution



Group Size



55%

Visitors are female

Slight female-skewed audience relevant for service tenant mix exhibit planning.

55%

Aged 30-39

Core target demographic for marketing & promotions.

57%

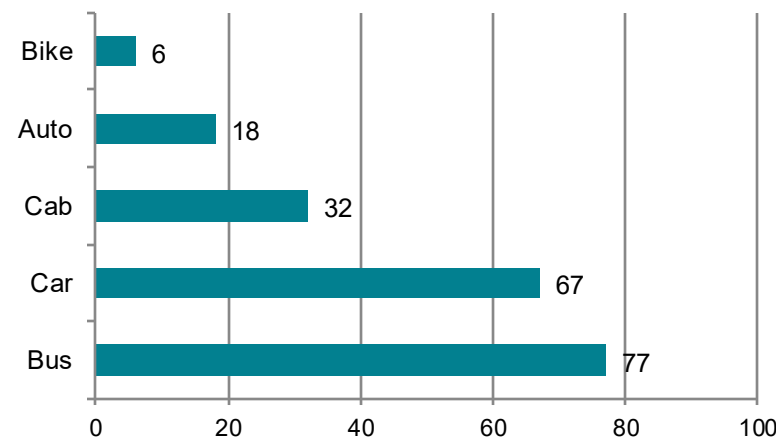
Groups of 3 or more

Mall is primarily visited as a social/family outing.

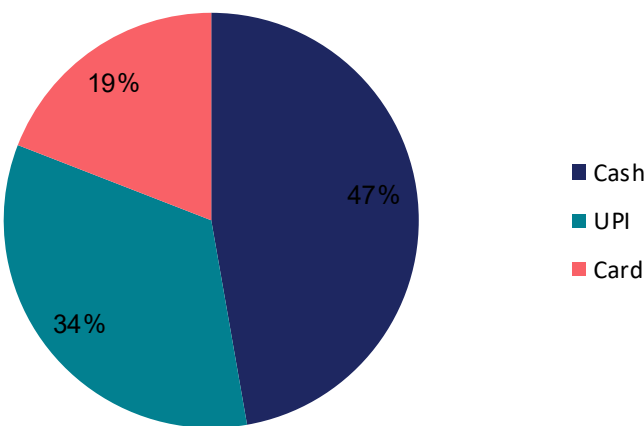
Inferences from Simulation (Part 2)

Operations: Transport, Payment & Services

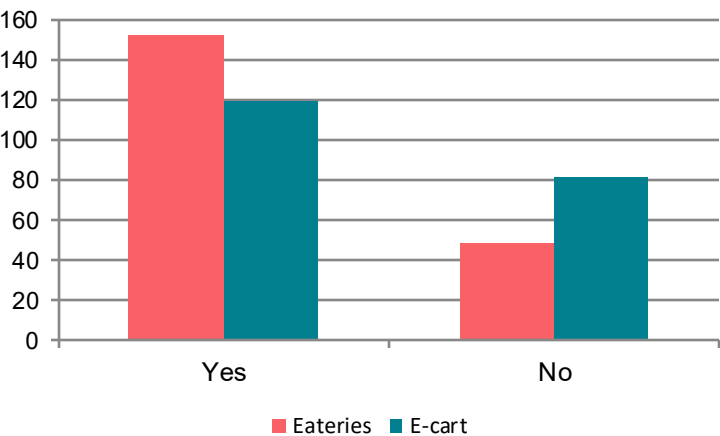
Mode of Transport



Payment Mode



Service Usage



72%

Public/shared transport

Bus + Cab + Auto dominate parking demand is moderate.

76%

Visit eateries

Strong F&B demand; on-site eateries are a critical revenue zone.

~50%

Visit > 90 minutes

Long dwell time opportunity for cross-category sales.

Conclusion

What the simulation tells us

Key Takeaway

The Monte Carlo simulation closely mirrors observed visitor behaviour at Jambu Zoo confirming that a young (30-39), socially-driven crowd dominates afternoon footfall, with a strong reliance on public transport, cash payments, and on-site eateries.



Marketing

Target ages 30–39 with weekend group offers (3+ persons).



Operations

Expand seating capacity kiosks



Services

Promote E-cart rental at entry gate only 60% currently aware/used.



Payments

Maintain robust Cash + UPI rails (84% of transactions).

Contribution of Each Member

Team responsibilities

1

Damia Sanjay Singh
20%

2

Vansh Amit Kotak
20%

3

E Sai Vihar
20%

4

Eshant Garg
20%

5

Kushagra
20%

— Thank You —